

House Price Prediction

Machine Learning — Task 2

Objective: Build a regression model to predict house prices using data preprocessing, feature engineering, model selection and evaluation.

1. Dataset & Initial Analysis

The dataset contains 4,600 records and 18 original columns. The target variable is **price**. Initial inspection showed no missing values. There were 49 records with zero price; these were removed, leaving 4,551 records.

2. Exploratory Data Analysis

The notebook includes the required four visualizations: scatter plot, box plot, histogram and pair plot.

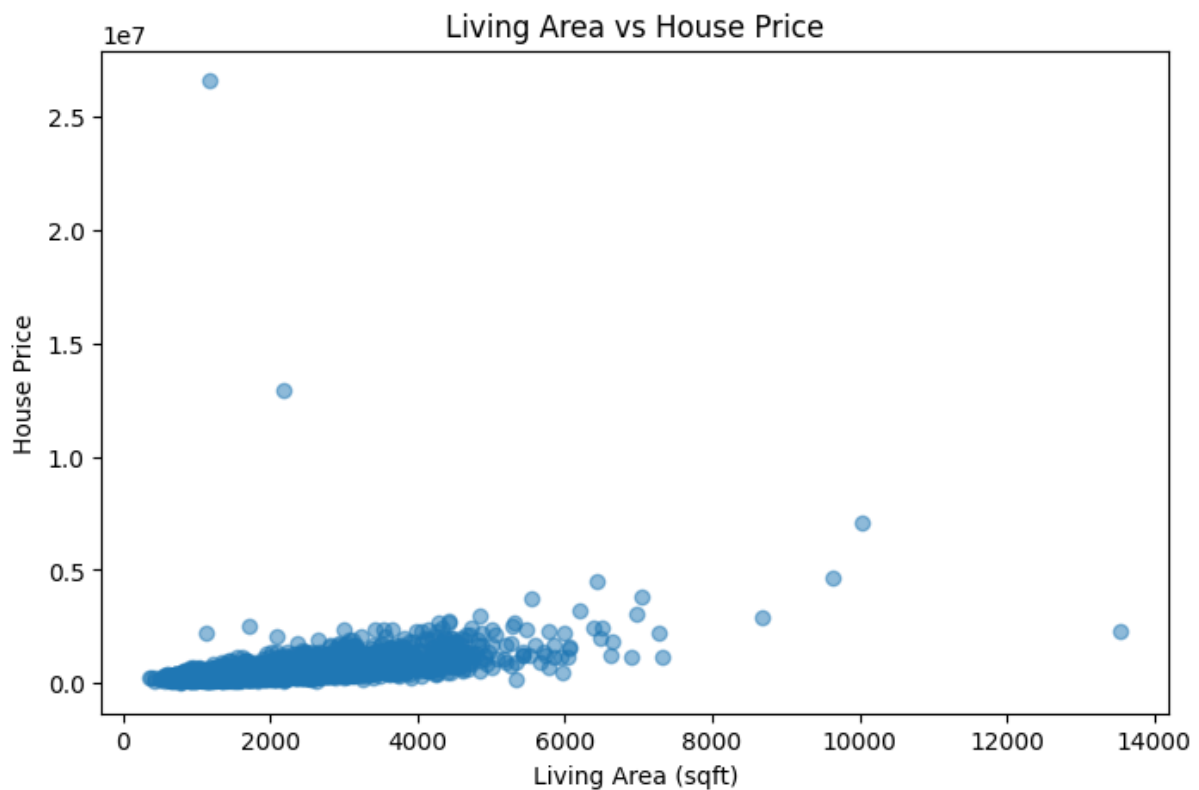


Figure: Scatter Plot

Box Plot of House Prices

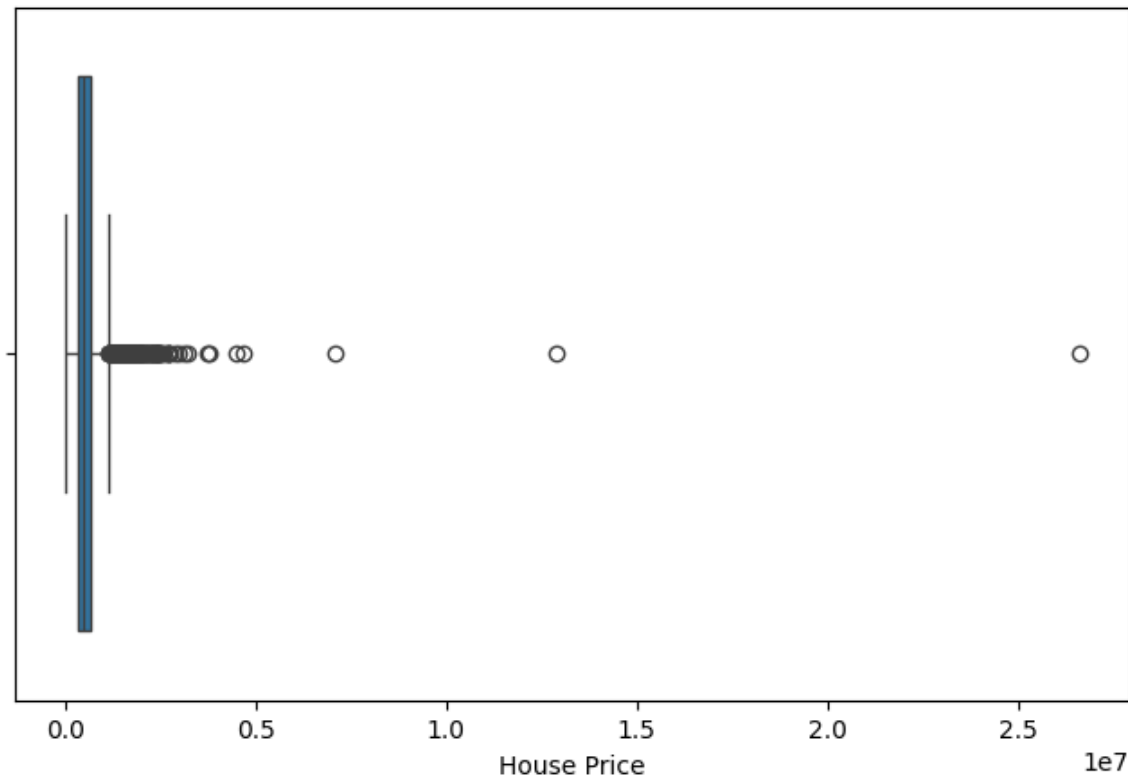


Figure: Box Plot

Distribution of House Prices

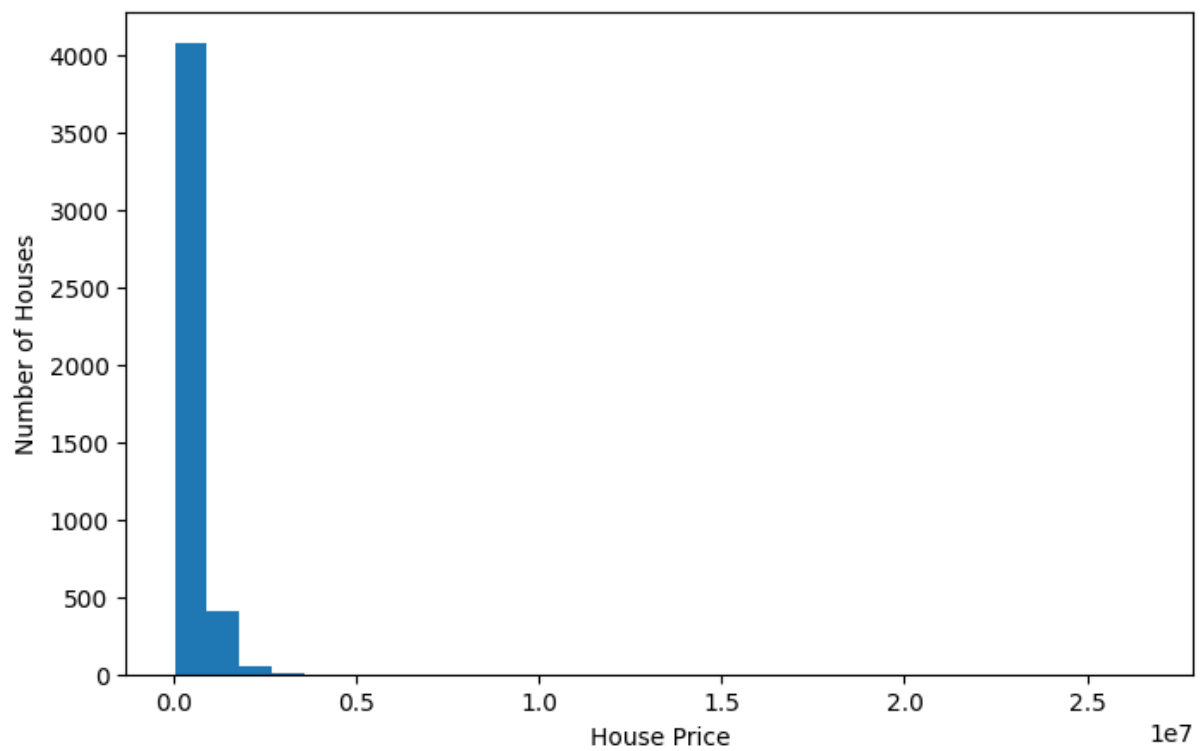


Figure: Histogram

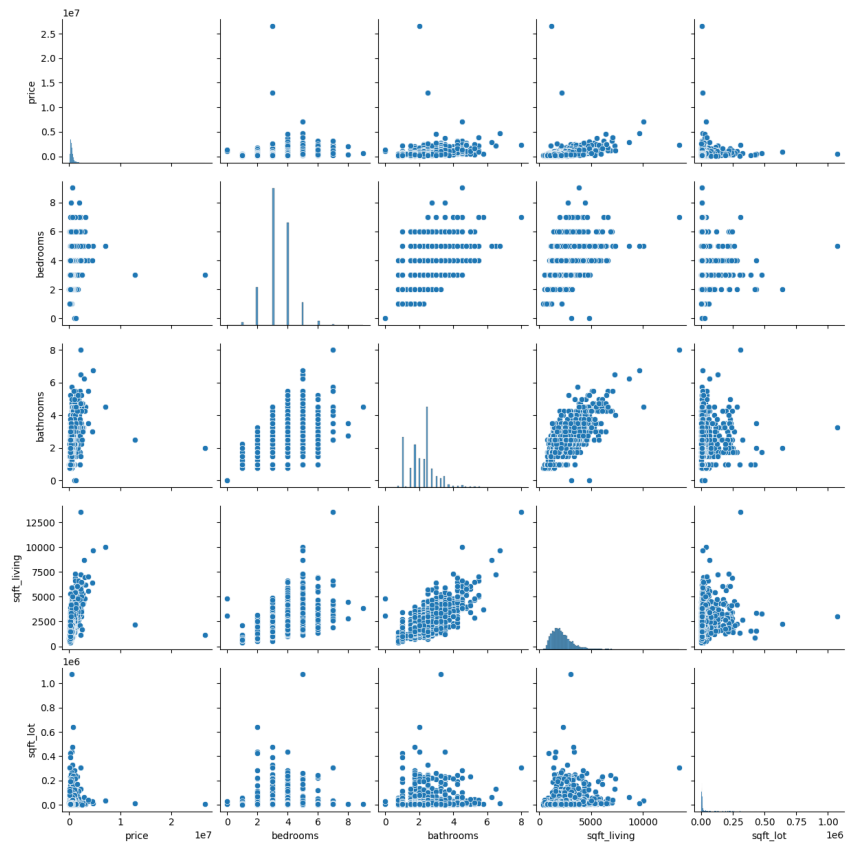


Figure: Pair Plot

3. Feature Engineering

The date column was transformed into **sale_year** and **sale_month**. New features **total_sqft** and **house_age** were created. The original date column was removed. High-cardinality **street** and constant **country** were removed. **city** and **statezip** were one-hot encoded.

Log transformation was applied to the target and selected numerical features. The modelling target was **log_price**.

4. Train-Test Split & Scaling

Training features: $3,640 \times 137$. Testing features: 911×137 . The target arrays contained 3,640 and 911 samples respectively. Feature scaling was completed before model training.

5. Model Comparison

Model	RMSE	MAE
Linear Regression	0.257282	0.154907
Gradient Boosting	0.285705	0.195852
Random Forest	0.287697	0.187354

6. Residual Analysis

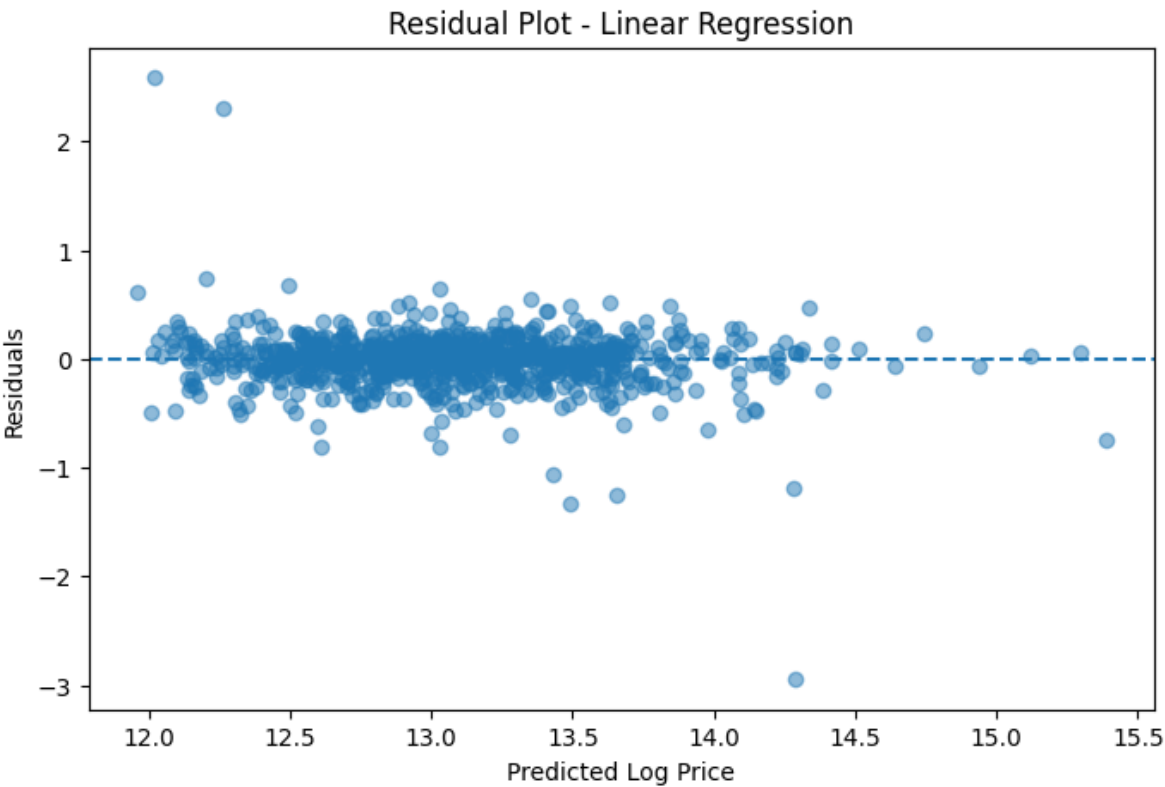


Figure: Residual analysis for Linear Regression.

7. Actual vs Predicted Price

The notebook includes an Actual vs Predicted analysis after converting predictions back from log-price to the original price scale.

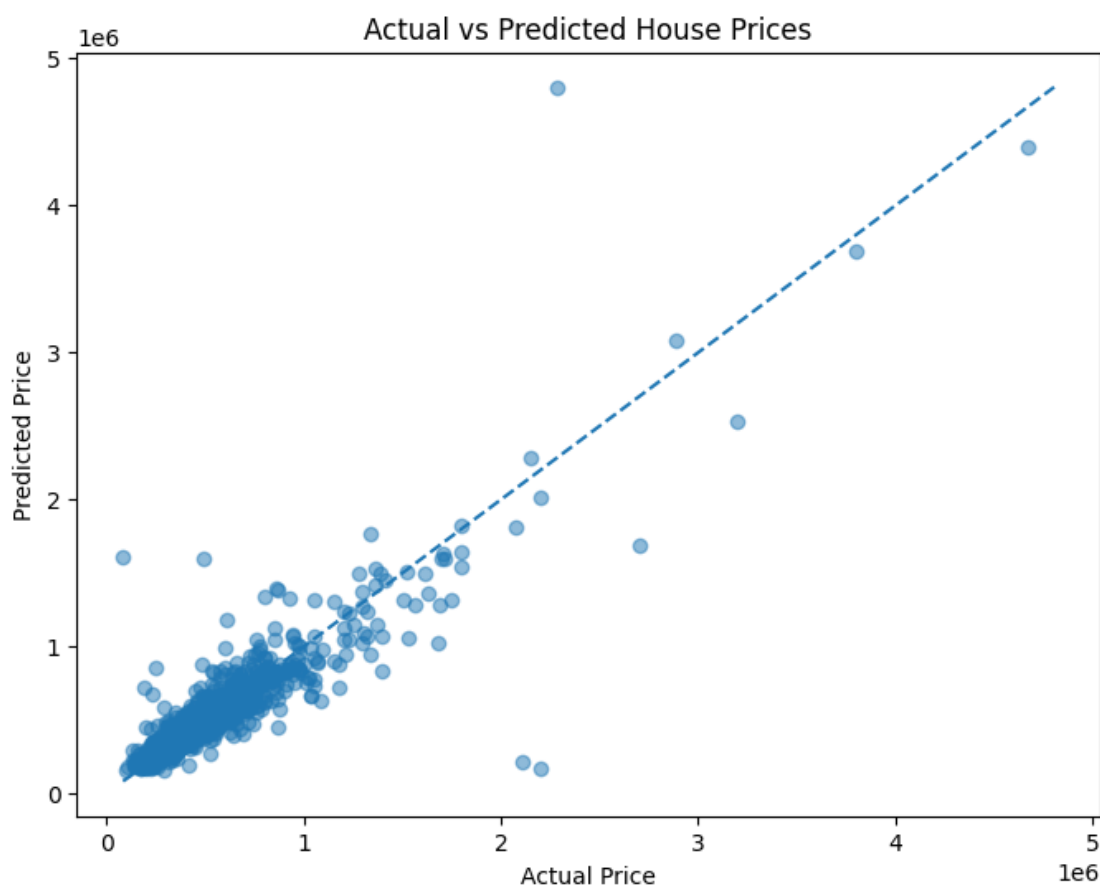


Figure: Actual vs Predicted House Price.

8. Sample Prediction

A sample prediction shown during the workflow was approximately **\$311,501.90**. The saved model was subsequently loaded successfully and tested.

9. Saved Artifacts

The project artifacts are **house_price_model.pkl**, **house_price_scaler.pkl**, and **requirements.txt**. The notebook contains the complete preprocessing, training and evaluation workflow.

10. Submission Links

Colab Notebook:

<https://colab.research.google.com/drive/1Zndt7b4kOB28QUYiS-XVZ6sx0r2YMHZP?usp=sharing>

GitHub Repository:

<https://github.com/RushikeshGame/House-Price-Prediction>

11. Conclusion

This project demonstrates a complete machine learning regression workflow covering EDA, preprocessing, feature engineering, encoding, log transformation, scaling, model comparison, RMSE/MAE evaluation, residual analysis, prediction and model saving.